the fluctuations gone, Einstein's laws describe the precise warping of space and time around black holes, and the precise slowing of time on Earth.

This is all the preamble to a punch line: *If Professor Brand could discover the quantum gravity laws for the bulk as well as our brane, then by discarding those laws' fluctuations, he could deduce the precise form of his equation (Chapter 25).* And that precise form would tell him the origin of the gravitational anomalies and how to control the anomalies—how to employ them (he hopes) to lift colonies off Earth.

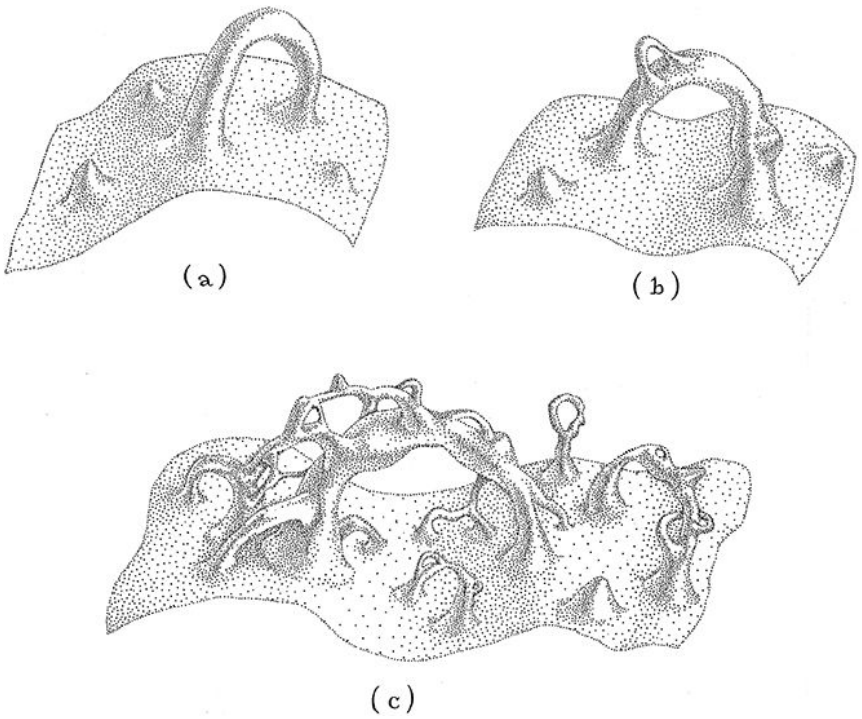


Fig. 26.3. Quantum foam. There is some probability (say, 0.4) that the foam will have the upper left shape, another probability (say, 0.5) for the upper right shape, and another (say, 0.1) for the lower shape. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy*.]*

In my extrapolation of the movie, the Professor knows this. And he also knows a place where the quantum gravity laws can be learned: inside *singularities*.